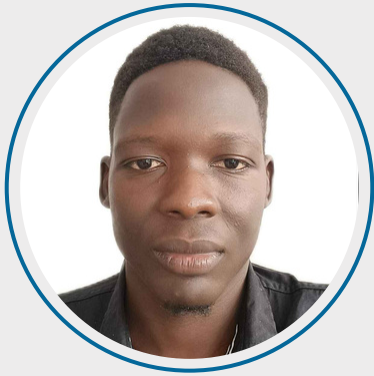




AMIR PETER PHILIP ADAM

Embedded Systems & IoT Engineer (Undergraduate)

CONTACT

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- www.amiradam.tech

SKILLS

TECHNICAL SKILLS

- Microcontrollers: ESP32, STM32, PIC18F
- Protocols: UART, SPI, I2C
- Tools: Keil, MPLAB, PlatformIO, EasyEDA
- Programming: C/C++, Python

SOFT SKILLS

- System-level debugging
- Effective Communication
- Critical Thinking
- Project Management

LANGUAGES

- English (Fluent)
- Arabic (Intermediate)

EDUCATION

- 2022 - Present**
CHANDIGARH UNIVERSITY
BE Electronics & Communications
- 2017 - 2021**
LOYOLA SECONDARY SCHOOL
General Sciences

PERSONAL PROFILE

Embedded Systems and IoT engineering student with hands-on experience in firmware development using C/C++ and Python on platforms including ESP32, STM32, and Raspberry Pi. Strong foundation in analogue and digital electronics, communication protocols, and system-level integration. Experienced in designing and implementing end-to-end embedded solutions involving sensor interfacing, data acquisition, and network communication. Actively seeking industrial exposure to apply engineering judgement, debugging skills, and practical problem-solving in real-world systems.

WORK EXPERIENCE

- DISASTER-RESILIENT OFFLINE MESH COMMUNICATION SYSTEM**
Tech Stack: ESP32, LoRa
 - Designed and implemented a disaster-resilient offline mesh communication system using ESP32 microcontrollers and long-range wireless communication (LoRa) to enable connectivity in network-isolated environments.
 - Developed peer-to-peer mesh networking to allow message transmission without reliance on cellular or internet infrastructure.
 - Integrated sensor data collection and local data processing to support situational awareness during emergency scenarios.
 - Optimized firmware for low-power operation to improve system reliability in constrained environments.
- END-TO-END IOT PIPELINE: ESP32 TO RASPBERRY PI WEB SERVER**
Tech Stack: ESP32, Raspberry Pi, Flask, HTTP, DHT11
 - Built an end-to-end IoT pipeline where ESP32 collected sensor data (temperature, humidity).
 - Transmitted data using HTTP POST to a Flask server hosted on Raspberry Pi.
 - Visualized real-time data via a custom web dashboard using HTML+CSS.
 - Implemented retry logic and communication handling to improve transmission reliability under unstable network conditions.
- HOME AUTOMATION USING ESP32 AND PYSERIAL**
Tech Stack: ESP32, Python
 - ESP32: Serves as the main controller that receives commands via serial and controls the lamps using relays.
 - Relays: The relays are used to switch the lamps on or off based on signals from the ESP32.
 - Pyserial: A Python script running on the computer sends control signals (on/off) to the ESP32 through the serial port.